Impacts of Hard Drug Use on HAART Efficacy in HIV-Positive Men

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Research Questions

- **How is the treatment response after 2 years of HAART affected by hard drug use?**
  - SF-36 Mental Component Summary (MCS) Score
  - SF-36 Physical Component Summary (PCS) Score
  - (log10) HIV viral load
  - CD4+ T cell count

- **Are any effects of hard drug use confounded or mediated by study protocol adherence?**

Data Overview

- 715 total subjects
- 467 subjects with all outcomes measured at Baseline and Year 2
  - 41% missing for drug users
  - 32% missing for non-users
- Adherence
  - Higher among hard drug users (n=1 with <95%)

Table 1: Descriptive Statistics of Baseline Covariates and Year 2 Adherence

Characteristic	No Hard Drug Use N = 437	Hard Drug Use N = 39	Overall N = 476
SF-36 MCS, mean (SD)	45.1 (13.7)	42.3 (11.2)	44.9 (13.5)
SF-36 PCS, mean (SD)	51.3 (9.1)	47.7 (8.5)	51.0 (9.1)
CD4+ T Cell Count, mean (SD)	375.4 (201.1)	352.2 (194.7)	373.5 (200.5)
Log Viral Load, mean (SD)	4.5 (0.9)	4.5 (0.9)	4.5 (0.9)
Age (Years), mean (SD)	43.1 (8.7)	44.6 (9.5)	43.3 (8.7)
BMI, mean (SD)	25.3 (4.4)	23.6 (3.4)	25.2 (4.3)
Missing/Improbable, n (%)	10 (2.3%)	3 (7.7%)	13 (2.7%)
Adherence, n (%)			
>=95%	388 (89%)	38 (97%)	426 (89%)
<95%	49 (11%)	1 (2.6%)	50 (11%)
Smoking Status, n (%)			
Never/Former Smoker	282 (65%)	9 (23%)	291 (61%)
Current Smoker	155 (35%)	30 (77%)	185 (39%)
Education, n (%)			
No College Degree	242 (55%)	29 (74%)	271 (57%)
College Degree or Greater	195 (45%)	10 (26%)	205 (43%)
Race/Ethnicity, n (%)			
Non-Hispanic White	279 (64%)	19 (49%)	298 (63%)
Other	158 (36%)	20 (51%)	178 (37%)

Analysis Methods

- Model Approach
  - Multiple Linear Regression
    - Bayesian and frequentist (OLS)
    - Outcome: Year 2 – Baseline
    - Adjusted for baseline measurement and covariates
      - Age, BMI, Smoking, Education, Race/Ethnicity
  - Clinically Significant Thresholds
    - MCS – 2 units
    - PCS – 2 units
    - log10 HIV viral load – 0.5 units
    - CD4+ T cell count – 50 $cells/mm^3$

- Model Summaries
  - Frequentist
    - Point Estimates
    - 95% Confidence Intervals (CI)
    - p-values
  - Bayesian
    - Point Estimates
    - 95% Highest Density Intervals (HDI)
    - Posterior Probabilities (PP)
    - LOO-IC

Results

- CD4+ Counts
  - Hard drug users experienced a lower treatment response
  - Estimated difference: -148.99 cells/mm^3; 95% HDI: [-206.12, -85.85]; PP: 0.999
- PCS
  - Hard drug users had a lower treatment response
  - Estimated difference: -3.33 units; 95% HDI: [-5.99, -0.66]; PP: 0.834
- MCS & Viral Load
  - No differences in treatment response between hard drug users and non-hard drug users

Table 2: Frequentist and Bayesian Estimates for Hard Drug Use Effects

	Frequentist			Bayesian			
Outcome	Estimate	95% CI	p-value	Estimate	95% HDI	Posterior Probability	Δ LOO-IC
SF-36 MCS	-0.29	[-3.75, 3.17]	0.870	-0.30	[-3.57, 3.32]	0.272	-3.51
SF-36 PCS	-3.34	[-6.06, -0.61]	0.017	-3.33	[-5.99, -0.66]	0.834	3.19
CD4+ Count	-163.88	[-226.35, -101.41]	<0.001	-148.99	[-206.12, -85.85]	0.999	24.31
Log Viral Load	-0.06	[-0.46, 0.34]	0.770	-0.05	[-0.44, 0.33]	0.018	-1.94

Note:
The posterior probability represents the probability of a clinically meaningful effect or greater.

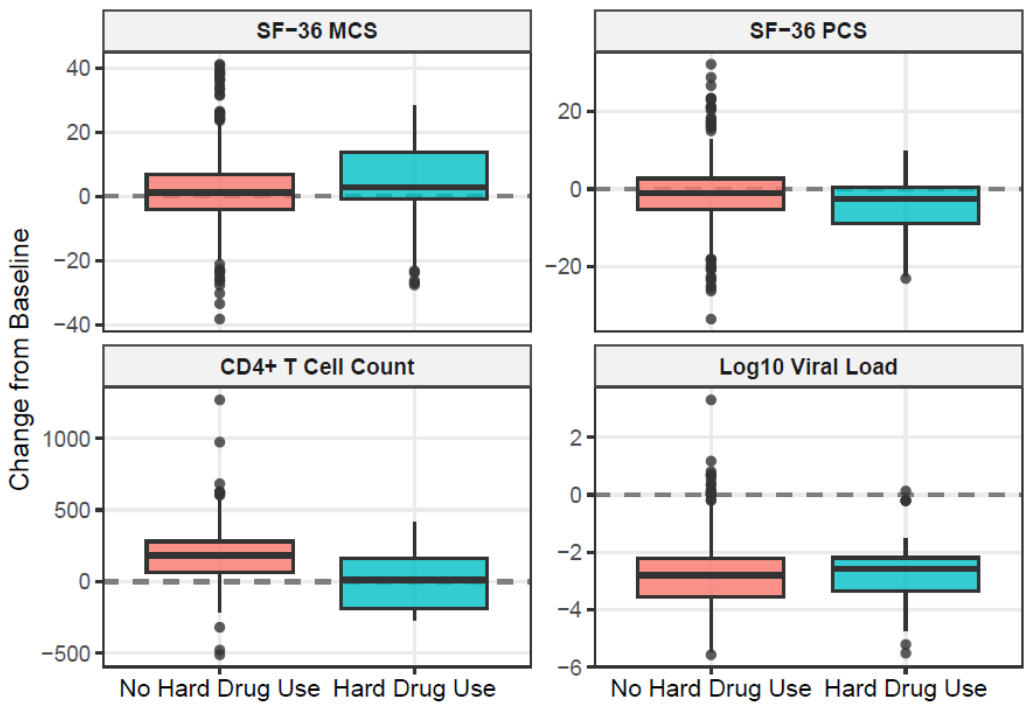

Figure 1: Treatment Responses Across Hard Drug Use Status

Results

- Adherence
  - No meaningful changes in hard drug use estimates after adjusting for adherence
    - 7.2% change in PCS
    - 3.5% change in CD4+ count
    - MCS and Log viral load stayed near 0
- Bayesian vs. frequentist
  - Consistent results between frameworks
  - Slight difference in CD4+ count estimates
    - -148.99 (Bayes) vs. -163.88 (freq)
    - Shrinkage due to $N(0, 100^2)$ prior

Table 3: Bayesian Estimates: Impact of Adjusting for Adherence on the Effect of Baseline Hard Drug Use (HDU)

	Unadjusted HDU Effect		Adjusted HDU Effect		Adherence Effect		
Outcome	Estimate	95% HDI	Estimate	95% HDI	Estimate	95% HDI	Δ LOO-IC
SF-36 MCS	-0.30	[-3.57, 3.32]	-0.54	[-3.71, 2.80]	-2.33	[-5.29, 0.64]	0.6
SF-36 PCS	-3.33	[-5.99, -0.66]	-3.57	[-6.38, -1.00]	-2.19	[-4.63, 0.22]	1.7
CD4+ T Cell Count	-148.99	[-206.12, -85.85]	-153.93	[-213.46, -91.22]	-50.35	[-105.77, -1.21]	1.7
Log Viral Load	-0.05	[-0.44, 0.33]	-0.01	[-0.40, 0.41]	0.44	[0.09, 0.78]	4.1

Table 4: Frequentist Estimates: Impact of Adjusting for Adherence on the Effect of Baseline Hard Drug Use (HDU)

	Unadjusted HDU Effect		Adjusted HDU Effect		Adherence Effect		
Outcome	Estimate	95% CI	Estimate	95% CI	Estimate	95% CI	p-value
SF-36 MCS	-0.29	[-3.75, 3.17]	-0.53	[-4.00, 2.94]	-2.30	[-5.35, 0.74]	0.137
SF-36 PCS	-3.34	[-6.06, -0.61]	-3.57	[-6.30, -0.84]	-2.20	[-4.59, 0.20]	0.072
CD4+ T Cell Count	-163.88	[-226.35, -101.41]	-169.68	[-232.21, -107.15]	-55.37	[-110.32, -0.42]	0.048
Log Viral Load	-0.06	[-0.46, 0.34]	-0.01	[-0.41, 0.38]	0.44	[0.09, 0.79]	0.013

Discussion & Limitations

- Hard drug use did not affect HAART response in viral load or MCS
- Less improvement in physical quality of life after HAART for hard drug users
- CD4+ Count treatment response severely diminished for hard drug users

- No evidence of mediation or confounding by adherence

- $N(0, 100^2)$ prior may be too informative for CD4+ Counts

- Limitations
  - Only 1 hard drug user with <95% adherence
  - Higher missingness among hard drug users (MAR)